The FCC should look at other ways to facilitate underlay of unlicensed communications in existing spectrum bands. Underlay can be achieved either through weak signals or adaptive, agile receivers. As technology advances, the FCC could consider a rule allowing underlay in certain bands, so long as devices check the local environment before transmitting and vacate a frequency within a certain number of milliseconds if a licensed service appears there. Underlay could also be used as a transition mechanism in bands where there are limited numbers of incumbents. Those incumbents could be allowed to remain in the band, but without the current guarantees against interference.

4. Drive Technology Development And Adoption

The government should seek out additional mechanisms to encourage the development and deployment of unlicensed devices. These could include liberalizing rules for experimental licenses, funding research projects, and using government procurement power to drive adoption of WiFi or other technologies.

The Defense Advanced Research Projects Agency (DARPA) of the Department of Defense has a distinguished history of supporting cutting-edge research in data networking, including the packet-switching technology that led to the Internet. DARPA has long funded research into meshed wireless networking, ultra-wideband, and software-defined radio, because of their military applications. These efforts should be continued, and every effort made to ensure smooth transfer of the resulting technologies to civilian applications.

The FCC and other relevant agencies should review their rules to identify unnecessary restrictions that keep unlicensed devices out of existing programs. For example, the FCC doesn't allow the use of Schools and Libraries subsidies for unlicensed networking devices, because they do not involve a communications "service." Of course, government should not try to pick winners among competitors in the marketplace, but rather work